Sharif Uddin

Electrical Engineering | RF, Microwave & Antenna Research

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Research Interests

RF and Microwave Engineering; Antenna Design and Characterization; Circularly Polarized Antennas; Full-Duplex and MIMO Wireless Systems; Relay Applications; mmWave and THz Antennas; Reconfigurable Antennas; Antennas-on-Chip; RF Integrated Circuits.

Education

Master of Science in Electrical Engineering [Fall 2025–Present]

Washington State University Vancouver, USA *GPA: [3.93/4.00]*

- Year of completion: May 2027
- Thesis: Wideband circularly polarized antennas for full-duplex wireless communication and relay application.
- Advisor: Dr. Tutku Karacolak

Master of Science in Electrical, Electronic and Communication Engineering – Only Coursework Completed [2022]–[2024]

Military Institute of Science and Technology (MIST), Bangladesh *CGPA: 3.58/4.00*

- Coursework completed; degree not awarded

B.Sc. in Electrical & Electronic Engineering [2016]–[2021]

Rajshahi University of Engineering & Technology (RUET), Bangladesh *CGPA: 3.54/4.00*

Research Experience

Graduate Researcher Fall 2025–Present

Washington State University, USA

- Design and investigate **differentially fed wideband circularly polarized antennas** for full-duplex & wireless relay communication systems.
- Perform full-wave electromagnetic simulations using **ANSYS HFSS**, followed by **antenna fabrication** and experimental validation.
- Characterize fabricated antennas in **anechoic chamber** and microwave measurement using **vector network analyzer**.

Selected Research Projects

Differentially Fed Wideband Circularly Polarized Antenna for Full-Duplex Communication

- Investigating wideband CP antenna architectures for simultaneous transmit and receive operation.
- Investigated polarization, spatial, structural, and decoupling-based isolation techniques to suppress inter-port coupling in full-duplex antenna systems.
- Characterized fabricated antenna prototypes through experimental measurement of differential S-parameters, radiation metrics, and key MIMO performance indicators (ECC, DG, and TARC).

Antenna Design, Simulation, Fabrication, and Measurement of Circularly Polarized Relay Antenna

- Designed and optimized wideband and circularly polarized relay antennas using full-wave electromagnetic simulation.
- Fabricated antenna prototypes and performed experimental characterization using an anechoic chamber.

Compact Wideband Circularly Polarized MIMO Antenna for High Data Rate Communications

- Designed and fabricated a compact C-shaped monopole circularly polarized antenna on FR4 using ANSYS HFSS and an LPKF laser milling machine for C- and X-band wireless communications after an intense parametric sweep.
- Upgraded the design to MIMO and evaluated the performance through channel capacity loss, mean effective gain, and validated circular polarization using surface-current and electric-field analyses.

Wideband Circularly Polarized Antenna Link - Effects on Transmission Loss and Gain

- Characterized antenna transmission loss and gain using a Keysight FieldFox N9952A 50 GHz microwave analyzer, applying Friis transmission equation for link-gain estimation.
- Applied Design of Experiments (DOE) and ANOVA to evaluate the effects of cable length, antenna separation, and measurement environment on transmission loss and gain, including uncertainty and calibration analysis.

Publications

Peer-Reviewed Publications

[1] **S. Uddin**, et al., "A Compact Wideband Circularly Polarized MIMO Antenna for High Data Rate Communications," *Presented at 2026 United States National Committee of URSI National Radio Science Meeting (USNC-URSI NRSM), Boulder, CO, USA,*

2026, pp. 457-457, DOI: https://doi.org/10.23919/NRSM68586.2026.11550840.

[2] **S. Uddin**, et al., "An Improved Switching Control Technique of a Three-Phase NPC Multilevel Converter in Reducing the Neutral Point Potential Swing," *Presented at the 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), Dhaka, Bangladesh, pp. 1439-1444, DOI: https://doi.org/10.1109/ICEEICT62016.2024.10534411.*

[3] A. Hossain, **S. Uddin**, et al., "Wavelet and Spectral Analysis of Normal and Abnormal Heart Sound for Diagnosing Cardiac Disorders," *BioMed Research International, 9092346, 16 pages, 2022. DOI: https://doi.org/10.1155/2022/9092346.*

Manuscripts Under Review

[1] **S. Uddin**, et al., "Differentially Fed Wideband Circularly Polarized Antenna for Full-Duplex Communications," manuscript submitted for publication, 2026.

Teaching Experience

Graduate Teaching Assistant, Department of Electrical & Computer Engineering [2025]–[Present]
Washington State University Vancouver, USA

- Grading and providing feedback on homework assignments and lab reports, and facilitating laboratory sessions for courses including Antenna Design and Analysis, RF Devices and Circuits, Electromagnetic Fields and Waves, and Principles of Solid-State Devices.

Lecturer, Department of Electrical & Electronic Engineering [2023]–[2025]
Northern University Bangladesh, Bangladesh

- Taught undergraduate courses in Electrical and Electronic Engineering and supervised undergraduate capstone projects, including poster presentations.

Lecturer, Department of Electrical & Electronic Engineering [2021]–[2023]
Prime University, Bangladesh

- Contributed to undergraduate teaching, laboratory instruction, student mentoring, and academic and extracurricular activities, including serving as sports advisor.

Technical Skills

Electromagnetics	Antenna design on rigid substrate, RF/microwave engineering, Circularly polarized antennas, MIMO, full-duplex systems, mmWave/THz antennas
Simulation	ANSYS HFSS, CST Studio Suite, Keysight ADS
Measurement	Using Anechoic Chamber, FieldFox Microwave Analyzer N9952A 50 GHz
Fabrication	LPKF laser & electronics milling antenna prototyping, Microfabrication: Thermal Oxidation, Photolithography, Sputtering, Lift-off, Etching, and Thin-film deposition
Programming & Data Analysis	MATLAB, Python/Google Colab, C/C++, Machine learning, Design of experiments (DOE), ANOVA.

Industry & Professional Experience

Industrial Visit — Pulse Larsen Antennas, YAGEO Group, Vancouver, WA, USA. [2026]

- Gained practical exposure to antenna design, testing, and characterization for real-world RF and wireless communication applications.
- Used antenna measurements in an advanced anechoic chamber and learned about practical techniques for evaluating performance.

Industrial tour at Kaptai 230 MW Hydroelectric Power Station, Chittagong, Bangladesh [2023]

- Gained practical exposure to 230 MW hydroelectric power generation, plant operations, and high-voltage electrical equipment maintenance from the biggest lake reservoir in Bangladesh.

Industrial attachment at Katakhali Peaking Power Plant, Rajshahi, Bangladesh [2019]

- Completed industrial training covering power generation, distribution, power system protection, electrical equipment, and plant operation & maintenance.

Awards and Honors

- Recipient, Four-Year Vocational Scholarship, Rajshahi University of Engineering & Technology (RUET), 2016–2019.
- Recipient, Scholarship for Excellence in Higher Secondary Certificate (HSC) Examination, Bangladesh Education Board, 2015.

References

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